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Avalanche stress in localized GaN epitaxial p–n diodes on 200 mm silicon substrates☆

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ABSTRACT

In this study, we demonstrate avalanche breakdown in pseudo-vertical GaN power devices grown by selective area growth (SAG) on 200 mm Si substrates, offering a pathway to high performance with silicon fab compatibility and low cost compared to GaN-on-GaN devices. A 1.5×10^{16} cm^{-3} silicon-doped GaN drift layer enabled the fabrication of an 805 V pseudo-vertical p–n diode. Device stability was initially evaluated by performing temperature-dependent reverse bias measurements (298–373 K) alongside DC stress conducted near the avalanche threshold. In the absence of dedicated edge termination, the peak electric field concentrates at the p–n junction sidewalls, triggering impact ionization and avalanche breakdown. Extended stress induces hard breakdown—driven by thermo-mechanical degradation—confirmed by post-failure SEM analysis. TCAD simulations corroborate the role of sidewall field enhancement in both avalanche initiation and failure mechanisms.

1. Introduction

Gallium Nitride (GaN) has emerged as a key material for next-generation power electronic devices due to its ability to operate efficiently under high electric fields and switching frequencies. Owing to its wide bandgap, GaN exhibits a higher critical breakdown field, faster carrier transport, and lower switching losses than conventional silicon-based semiconductors, which enables superior performance in high-power and high-frequency operating regimes. These advantages have driven increased research and industrial adoption of GaN devices in application areas such as wireless communication systems, electric vehicles, and renewable energy power conversion technologies [1,2]. One notable technological progression in this field is the transition from conventional lateral GaN device architectures to vertical GaN structures. In vertical GaN devices, current flows normal to the substrate rather than along the surface, enabling more effective utilization of the device volume for voltage blocking [3]. This structural orientation supports higher current conduction capability and enhanced heat dissipation, which are critical requirements for high-voltage and high-power applications [3,4].

The availability of free-standing GaN substrates provides an attractive pathway toward low-strain GaN epitaxial growth and enables the realization of fully vertical device architectures [5]. Despite these advantages, their practical deployment remains limited due to several constraints, including high production costs, non-uniform material quality, restricted wafer diameters, and limited commercial supply. As a consequence, substantial research attention has shifted toward the development of vertical GaN power devices on foreign substrates, such as silicon, sapphire, and Qromis Substrate Technology (QST™) platforms [6].

Among these alternatives, silicon (Si) substrates are particularly appealing because of their low cost and compatibility with mature, large-diameter semiconductor manufacturing processes. However, the significant differences in lattice constants and thermal expansion coefficients between GaN and Si introduce considerable challenges during epitaxial growth. Although various strain management strategies have been employed to alleviate these issues, the overall epitaxial thickness achievable on Si substrates is typically restricted to approximately 10 μm, thereby limiting the thickness of the drift region required for high-voltage operation [7,8]. To address this constraint and enhance the

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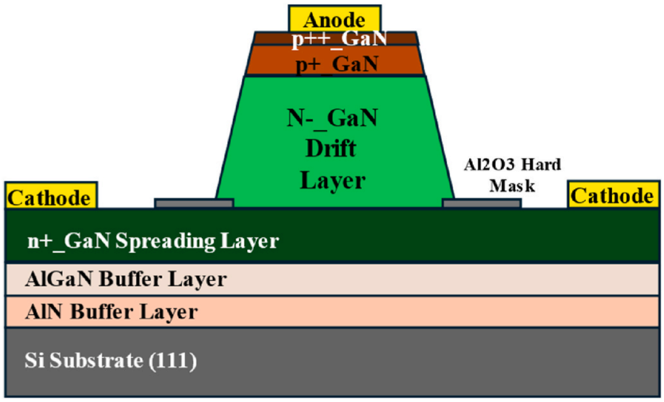

Fig. 1. Schematic 3D view of pseudo-vertical GaN p-n diode grown by SAG on Si substrate.

maximum attainable breakdown voltage (BV), selective-area growth (SAG) techniques on 200 mm Si wafers have been proposed. By promoting localized strain relaxation at the boundaries of patterned growth regions, SAG reduces thermally induced stress and enables the formation of stress-relieved GaN drift layers with thicknesses approaching 8 μm, which are well suited for vertical device implementations [9–11].

Although continuous progress has been achieved in the development of pseudo-vertical GaN-on-Si p–n diodes (PNDs), their avalanche breakdown behavior—an important indicator of device reliability—has not yet been fully clarified [12,13]. In particular, avalanche capability has not been conclusively demonstrated for devices incorporating SAG-grown GaN drift layers [14]. The ability to sustain controlled avalanche operation is critical for power electronic applications, as it ensures non-destructive behavior under near-breakdown conditions, thereby improving device robustness and overall system reliability. Nevertheless, several factors can hinder this capability, including parasitic leakage currents along mesa sidewalls, elevated threading dislocation densities within the GaN drift layer, and localized point defects, all of which may lead to premature or destructive breakdown events [15]. These challenges continue to represent significant obstacles to achieving reliable avalanche performance in GaN-on-silicon power devices [16]. In our recent studies, the avalanche characteristics of similar device structures were examined using unclamped inductive switching (UIS) measurement [17]. However, this characterization technique subjects devices to severe electrical and thermal stress, which can result in catastrophic failure and complicate the reliable assessment of intrinsic avalanche behavior.

This study reports avalanche operation up to 805 V in pseudo-vertical GaN p–n diodes fabricated by selective-area growth (SAG) on silicon substrates, together with comprehensive electrical characterization under both forward and reverse bias conditions. In contrast to our previous work [17], avalanche behavior is examined using a DC stress methodology, originally developed for SiC devices [16] and adapted here for GaN diodes. In this technique, the device is biased with a constant current near the avalanche threshold while the resulting voltage transients are monitored in real time. Supporting TCAD Synopsys simulations indicate pronounced electric field crowding at unpassivated mesa sidewalls, which correlates well with impact ionization locations identified through SEM analysis. Collectively, these findings indicate that SAG-based GaN-on-silicon platforms not only enable scalable vertical device architectures but also exhibit stable avalanche behavior, helping to close the performance gap between lateral GaN-on-Si HEMTs and emerging vertical GaN power device technologies.

2. Device structure and fabrication

Fig. 1 illustrates a 3D view of the pseudo-vertical p–n diode we

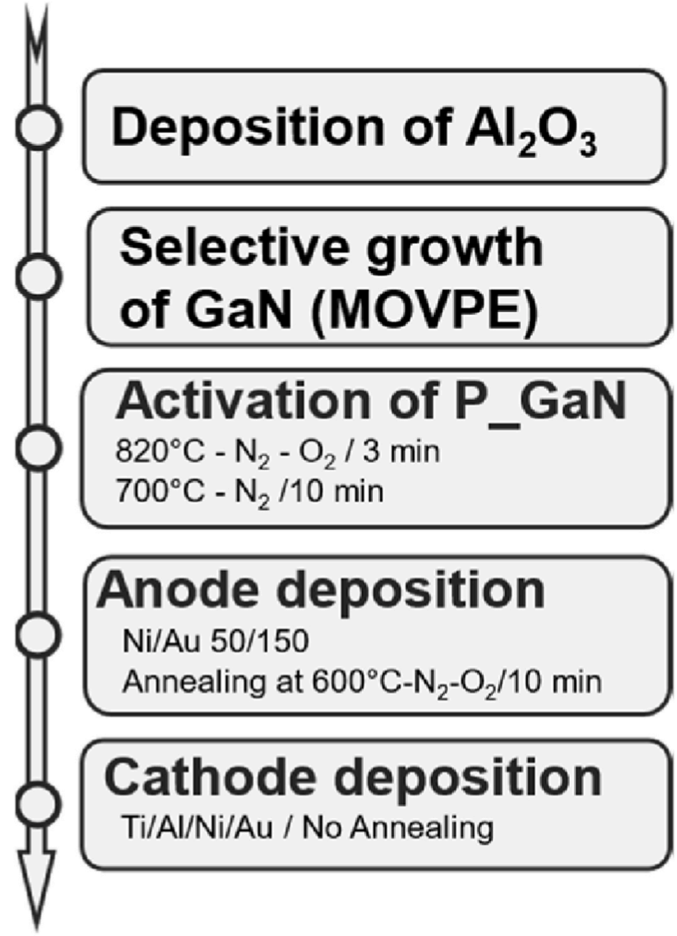

Fig. 2. The device fabrication steps of GaN pseudo-vertical p-n diode.

fabricated. The device starts on a 200 mm silicon (111) substrate, on which we grew the epitaxial stack using metal-organic vapor phase epitaxy (MOVPE). The stack includes AlGaN/AlN buffer layers, a 300 nm unintentionally doped GaN layer, and three n-type GaN layers, each 350 nm thick, with decreasing doping levels from top to bottom (5×10^{18}, 5×10^{17}, and 5×10^{16} cm^{-3}). These three layers were originally designed as calibration layers for cross-sectional resistance measurements; however, their thinness made accurate characterization challenging [10].

The fabrication process, shown in Fig. 2, begins with the deposition of a 50 nm Al_2O_3 hard mask by atomic layer deposition (ALD) at 300 °C. This mask is patterned to define openings for selective-area growth (SAG). Within these 100 μm diameter apertures, we grow an 8 μm-thick Si-doped n$^-$ GaN drift layer (targeted to be 1×10^{16} cm^{-3}), followed by a 0.8 μm p + -type GaN layer doped to (Cp_2Mg: 5×10^{18} cm^{-3}). And a 50 nm-thick p++-type Mg-doped capping layer with a doping concentration of 1×10^{19} cm^{-3}. The p + -GaN layer is then activated through a two-step annealing process: first at 820 °C for 3 min in an N_2/O_2 atmosphere, and then at 700 °C for 10 min in N_2. For the electrical contacts, a Ni/Au stack (50 nm/150 nm) is deposited on the p++-GaN to form the anode and alloyed at 600 °C for 10 min. The cathode is formed using a Ti/Al/Ni/Au stack (95 nm/200 nm/20 nm/165 nm), without any post-metallization annealing to preserve the integrity of the anode contact.

Fig. 3 (a) shows a top-view SEM image of the GaN mesas after growth, revealing their characteristic hexagonal shape. Fig. 3 (b) highlights a naturally formed bevel at the mesa sidewalls, approximately 65° in angle. This bevel arises spontaneously from the anisotropic growth of the wurtzite GaN crystal: different crystallographic planes grow at

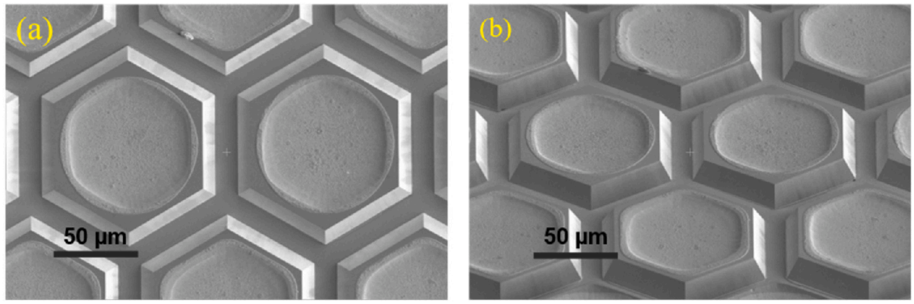

Fig. 3. SEM images of the hexagonal structures: (a) and (b) 30° tilted view showing 65° beveled sidewalls.

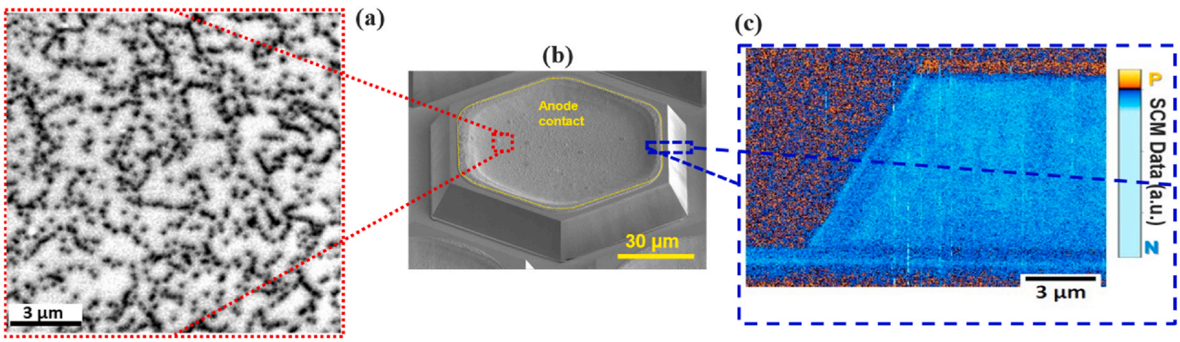

Fig. 4. (a) Cathodoluminescence (CL) plan-view image of the as-grown GaN layer on Si, (b) SEM images of the hexagonal mesa and (c) SCM measurements on the edge of the cross-section, reprinted from Ref. [14].

different rates, producing well-defined facets. For our devices grown along the c-plane, the sidewalls correspond to the $\{10\bar{1}1\}$ planes.

Interestingly, this naturally formed bevel plays an important role in improving device performance. Sharp p–n junction edges tend to concentrate electric fields, which can lead to premature breakdown or localized damage. Conventional approaches often introduce a controlled bevel via mesa etching to redistribute the electric field, but this extra processing can create surface defects and increase leakage currents. In our devices, the spontaneously formed bevel smooths the junction edges without any additional steps, reducing peak electric field intensity, enhancing reliability, and lowering the risk of edge-induced breakdown [18–20].

The threading dislocation density (TDD) was evaluated by cathodoluminescence (CL) imaging, as illustrated in Fig. 4 (a). Dislocations appear as non-emissive features due to their role as non-radiative recombination centers and were quantified by counting the corresponding dark contrasts. An exact determination is challenging, as individual dislocations can coalesce into continuous dark lines or clusters. Nevertheless, the TDD is estimated to be approximately $(2 \pm 1) \times 10^8$ cm^{-2}. Fig. 4 (c) presents scanning capacitance microscopy (SCM) data acquired near the cross-sectional edges, enabling visualization of the doped regions. The p-type GaN layer is represented by orange contrast, while the n-type drift region appears in blue. No p-GaN signal is detected at the island sidewalls, with p-type material observed only on the top facets, consistent with our previous report [14]. This behavior may result from limited magnesium incorporation on semi-polar surfaces, carrier depletion effects, or a p-GaN thickness below the spatial resolution of the SCM technique [21].

3. Device performance and discussion

3.1. Capacitance-voltage (C-V) measurements

Capacitance measurements were performed using a Keysight (Agilent) B1520A Multi-Frequency Capacitance Measurement Unit (MFCMU). As shown in Fig. 5, from high frequency C-V measurements at 100 kHz, using the equation presented in the figure [22], the net doping concentrations N_D-N_A was extracted, giving ~ 1.5 x 10^{16} cm^{-3}; where N_D, N_A, q, ε_s, ε_0 and A denote the donor and acceptor concentration in n− GaN, the electron charge, the relative permittivity of GaN, the vacuum permittivity and the area of the anode, respectively. This value is a bit higher than the intended one (1×10^{16} cm^{-3}), which may result from unintentional background donors (e.g., oxygen or nitrogen vacancies) and possible overestimation inherent to C–V profiling due to deep levels or junction non-idealities.

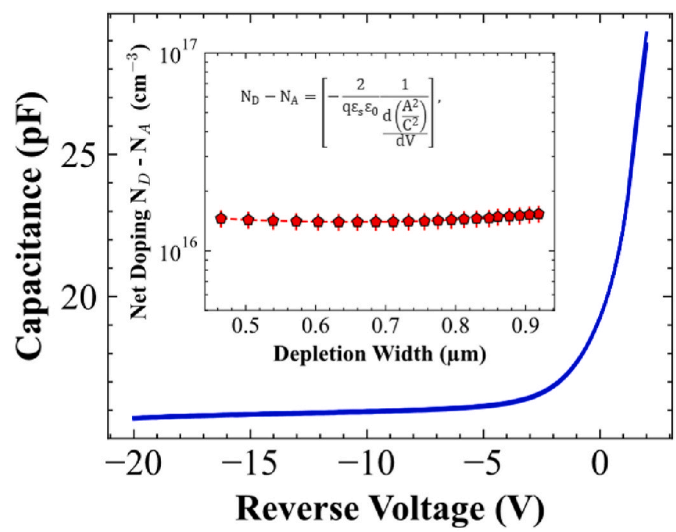

Fig. 5. Typical Capacitance-Voltage measurement. Inset: Net doping profile extracted using the capacitance-voltage (C–V) method, showing the variation of effective doping N_D−N_A as a function of depletion width.

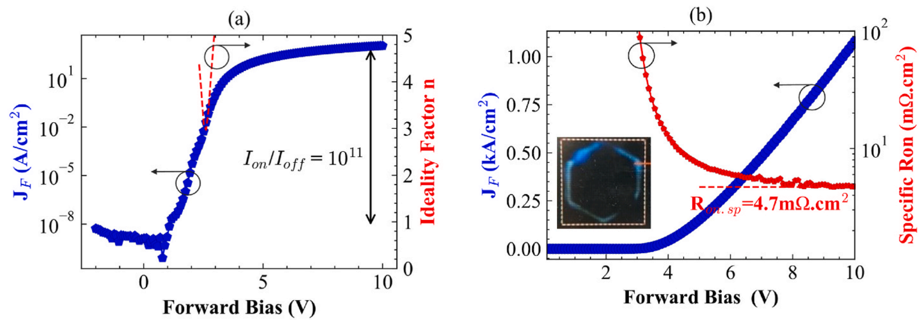

Fig. 6. Representative forward current–voltage (J_F–V) characteristics of the diode: (a) semi-logarithmic scale, showing the extracted ideality factor (red, right axis), and (b) linear scale, showing the extracted specific on-resistance $R_{on,sp}$ (red, right axis) (Inset: optical microscopy image of the electroluminescence.). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

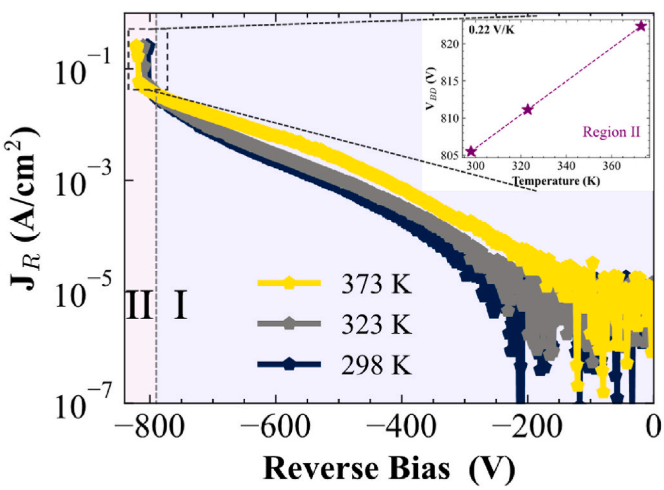

Fig. 7. Typical reverse characteristics of PV p-n diodes at 298 K, 323 K and 373 K. Inset: Enlarged view of the breakdown region above 790 V. Inset: Extracted breakdown voltage as a function of temperature, confirming avalanche behavior.

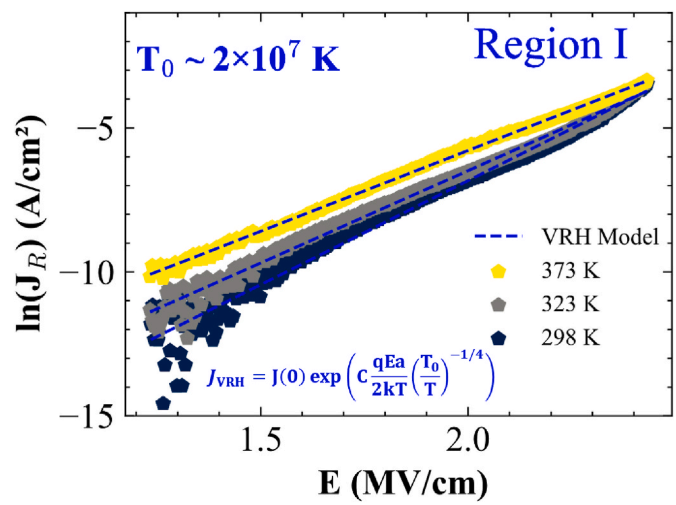

Fig. 8. Plot of ln (J_R) as a function of applied electric field E, fitted using the variable-range hopping (VRH) model over a voltage range from 0 to −600 V, with a characteristic temperature T_0 of 2.9 × 10^7 K—within the typical range of 10^6–10^9 K [26,27].

3.2. Forward I-V measurements

Forward I–V measurements of the p–n diodes were carried out by sweeping the bias voltage between the anode and cathode using a Keysight (Agilent) 4156C precision semiconductor parameter analyzer. Fig. 6 (a) and (b) present the forward J_F–V characteristics of the pseudo-vertical PND, recorded at room temperature and shown in both semi-logarithmic and linear scales. The device reaches a current density of 1.2 kA/cm^2 at 10 V, with a specific on-resistance (Ron,sp) of 4.7 mΩ cm^2. It also demonstrates a favorable turn-on voltage of 3.1 V, measured at a current density of 1 A/cm^2, along with an impressive on/off current ratio of 10^{11}. The reported Ron,sp is somewhat overestimated, however, due to additional series resistance from the relatively large contact separation (over 300 μm) resulting from a non-optimized device layout [23]. From the forward I–V analysis, the ideality factor was found to be approximately 3. This relatively high value suggests that multiple recombination mechanisms—such as radiative recombination, evidenced by the luminescence seen in the inset of Fig. 6 (b), and Shockley-Read-Hall (SRH) recombination—play a role in carrier transport [24].

3.3. Reverse I-V measurements

Fig. 7 shows the temperature-dependent reverse J_R–V–T characteristics of the GaN pseudo-vertical PND, measured over a temperature range of 298 K to 373 K. At room temperature, the device exhibits a nondestructive breakdown voltage (BV) of approximately 805 V. To prevent arcing with air during the measurements, the devices were immersed in Fluorinert. This performance highlights the critical role of selective area growth (SAG) in producing a high-quality GaN drift layer with naturally beveled mesa edges, eliminating the need for etching and avoiding etch-induced damage [25]. The diodes achieve a breakdown voltage that approaches 75 % of the theoretical parallel-plane maximum of 1060 V.

The reverse-bias leakage current was also analyzed using the temperature-dependent J_R–V measurements [26]. These measurements were performed on the PND up to −805 V across a temperature range of 298 K to 373 K, as illustrated in Fig. 7.

The reverse-bias characteristics can be clearly divided into two distinct voltage regions, marked by “I” and “II.” In region I, from 0 to −790 V, the leakage current is primarily driven by variable-range hopping (VRH), as described in our previous study [26]. As the electric field strengthens, electrons move from the valence band to the conduction band via hopping along threading dislocations, making VRH the dominant transport mechanism across all measured temperatures. This interpretation is supported by the observed linear trend between ln (J) and the electric field, E, shown in Fig. 8. Additionally, the extracted characteristic temperature, T_0, is 2 ± 2 × 10^7 K, which falls within the typical range of 10^6–10^9 K and aligns well with reported values in the literature (4.92 × 10^7 K for ND ~3 × 10^{16} cm^{-3}) [26,27].

In region “II,” at voltages above −800 V, the leakage current rises

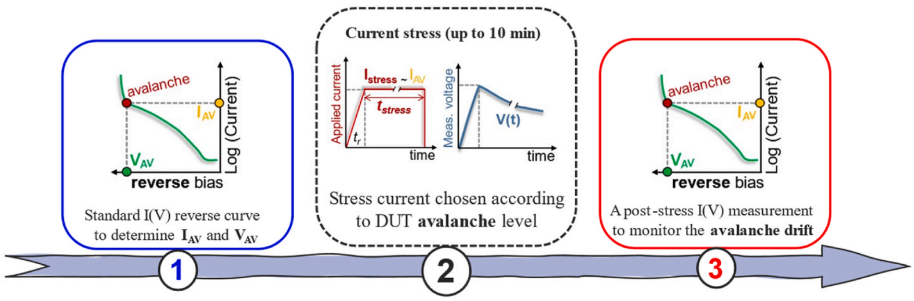

Fig. 9. Test sequence for the proposed method. Step 1 is a standard reverse I(V) measurement. Step 2 is a DC current stress using a constant I_{stress} closed to avalanche. Additional static I(V) (step 3) enable to monitor I_{AV} and V_{AV} potential drift [32].

noticeably without causing permanent damage to the device, indicating that localized epitaxy may support avalanche operation—a key feature for device reliability. In this range, avalanche breakdown acts as a controlled leakage mechanism, allowing the device to safely dissipate excess carriers through impact ionization and avoid irreversible failure [22]. As the temperature increases, stronger phonon scattering delays the onset of impact ionization, meaning higher voltages are required for carriers to reach the necessary kinetic energy. This effect is reflected in the positive temperature dependence of the breakdown voltage, shown in the inset of Fig. 7. The positive temperature dependence of the breakdown voltage, with an estimated coefficient of 0.22 V/K [22], provides clear evidence of the device's avalanche capability.

The devices exhibit a Baliga figure of merit (BFOM) of 0.13 GW/cm^2, which is comparable to values reported for devices grown using conventional epitaxial methods [19,25,28], demonstrating that selective area growth (SAG) can produce high-quality GaN layers. However, this value is still lower than some literature reports, likely due to the relatively high on-resistance in our devices. This emphasizes the need for further optimization to reduce the on-resistance and fully realize the potential performance of these diodes.

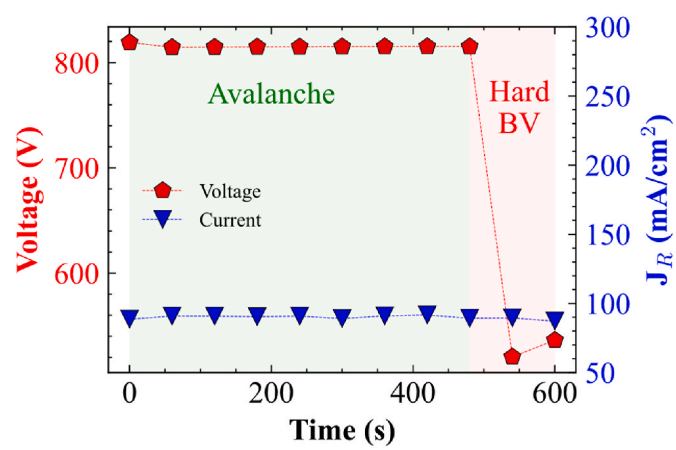

Fig. 10. Voltage and current stress during Step 2 as a function of time, with stress applied for up to 10 min. The device operates in avalanche mode up to 7 min, followed by hard breakdown at higher voltage beyond this duration.

3.4. DC stress current near to avalanche mode

Evaluating the reverse robustness of power devices is critical, especially for applications involving high voltage or harsh operating environments. Depending on the device maturity and the targeted reliability standards, various electrical stress methodologies can be employed [29]. Such as HTRB, H3TRB[30] and UIS [31].

In this study, two methods were considered: unclamped inductive switching (UIS) and DC avalanche stress. Here, we report only the latter, while the UIS results will be presented in a forthcoming publication.

The DC stress (in avalanche mode) methodology proposed here was recently developed for SiC devices [16]. It involves exposing the devices to controlled avalanche conditions for different time intervals, followed by an analysis of the resulting variations in key static electrical parameters—such as leakage current and avalanche voltage.

This method enables rapid, preliminary screening of new device designs or process variations—much faster than conventional reliability tests. Devices are first characterized directly on-wafer using only a wafer prober and a high-voltage source-measure unit (HVSMU), the setup configuration is discussed in details in Refs. [16,32]. The method proposed to evaluate the robustness and stability of GaN diodes under near-avalanche conditions consists of three primary stages, as shown in Fig. 9. This stepwise protocol enables a systematic and comprehensive assessment of the device response during both the application of near-avalanche stress and throughout the subsequent recovery period.

Step **1:** A standard reverse I–V measurement is first performed to establish the baseline electrical behavior of the device under test (DUT). For the GaN diodes studied, this measurement yields the avalanche onset current I_{AV} and provides key parameters such as reverse leakage current and breakdown voltage.

Step **2:** The DUT is then subjected to a DC stress slightly above the previously determined I_{AV} for a specified duration (up to 10 min). As illustrated in Fig. 9, stress may be implemented under either current or voltage control; both approaches produce comparable trends, although current-biasing is favored because it offers higher accuracy and a larger dynamic range above the avalanche point. The applied current magnitude and stress duration can be varied to probe different near-avalanche degradation scenarios.

Step **3:** Following the stress period, a second reverse I–V measurement is performed and compared with the initial baseline. This comparison reveals any drift in electrical parameters—e.g., increases in leakage current or shifts in avalanche voltage—which may signify degradation mechanisms or structural changes induced by the near-avalanche stress.

The main advantage of this method is that it activates the impact ionization of both electrons and holes under high electric field conditions, enabling the identification of potential instabilities or parameter drifts closely tied to the spatial location of avalanche breakdown [22].

Based on the previously discussed results (at 25 °C) presented in Fig. 7, the approximate avalanche voltage (V_{AV}) is determined to be around −818 V in addition, the stress current (J_{AV}) was set to 10 μA—slightly above the avalanche current—to provide a safety margin against measurement fluctuations while ensuring operation in the near-avalanche regime.

The DUT was then stressed at I_{AV} = 30 μA for 10 min while the avalanche voltage was monitored. Fig. 10 presents the time evolution of avalanche voltage and current during the stress period. Voltage

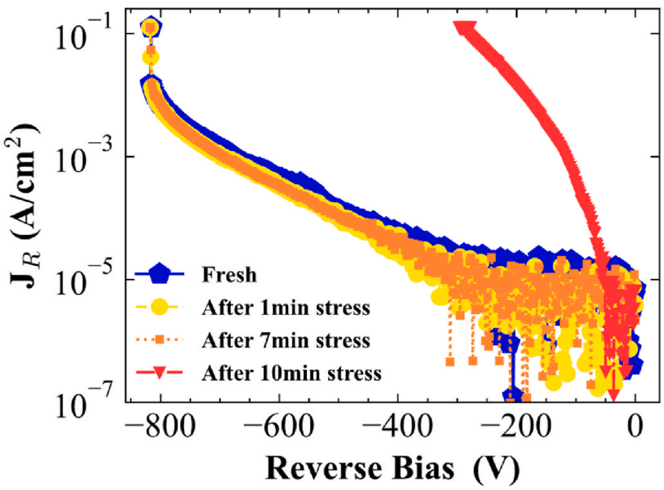

Fig. 11. Reverse J_R–V characteristics from Steps 1 and 3 after 10 min of current stress for a GaN pseudo-vertical p–n diode on a Si substrate.

monitoring began at t = 0 s after the ramp-up, with an initial avalanche voltage of approximately −820 V. The device maintained stable operation in avalanche mode for the first 7 min, after which a sudden voltage collapse signaled catastrophic failure resulting from stress-induced damage.

Fig. 11 shows the J_R(V) characteristics for an 805 V-rated p-n diode, measured before (step 1) and after (step 3) the application of a stress condition (step 2). The curve corresponding to step 1 (Fresh measurement) shows the initial behavior of the diode, while the curve for step 3 reveals the effects of the stress (After) at various durations (1, 7, and 10 min). It is observed that the reverse leakage current (J_R) in the pre-avalanche region remains essentially unchanged across all measurements, indicating that the applied stress does not affect leakage current behavior prior to avalanche onset. This stability suggests the absence of significant degradation mechanisms in the moderate-field regime. Moreover, the results show stable avalanche behavior up to 7 min, while a hard breakdown is observed after 10 min, likely caused by thermo-mechanical stress due to prolonged high-field and avalanche current exposure. In fact, the generation of electron–hole pairs during impact ionization can lead to localized current filaments and self-heating, which degrade carrier mobility and accelerate device degradation [31]. Additionally, the absence of passivation increases the likelihood of breakdown, as edge electric fields remain poorly managed.

To investigate the underlying failure mechanism, high-resolution SEM imaging was conducted on the n-GaN drift region after catastrophic breakdown. As shown in Fig. 12, several distinct forms of physical damage were observed, including localized melting of the surface, delamination of the anode metallization layer, and the formation of pits, with damage concentrated predominantly along the device edge. This spatial distribution of degradation is consistent with the presence of intense field crowding at the periphery due to the absence of effective edge termination. Without proper termination structures, the electric field lines become highly concentrated near the periphery, leading to localized current density peaks. Under sustained avalanche operation, these localized high-field regions promote enhanced impact ionization, generating large numbers of electron–hole pairs. The resulting carrier multiplication increases the local current density, which in turn leads to pronounced Joule heating [31]. This thermal build-up produces significant thermoelastic strain within both the metal contact and the GaN substrate, thereby weakening interfacial adhesion and triggering delamination. Simultaneously, thermal gradients and localized overheating induce plastic deformation and surface melting, while repeated thermal cycling during stress accelerates crack initiation and propagation.

In addition, TCAD Sentaurus Synopsis numerical device simulations corroborate these experimental findings. As presented in Fig. 13, and based on the findings reported in Ref. [33], the electron and hole

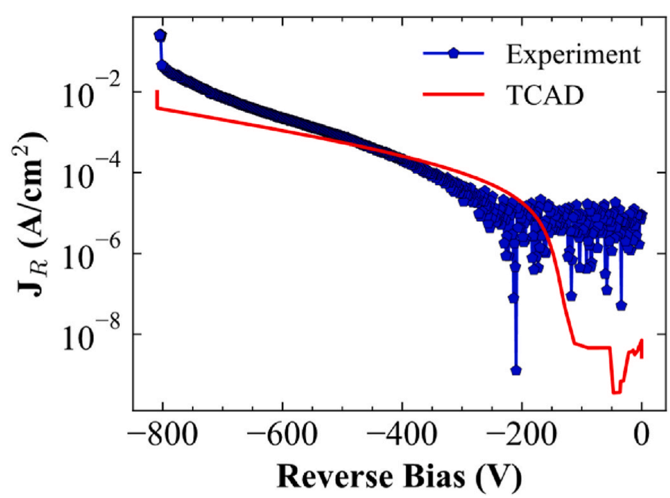

Fig. 13. Simulated avalanche breakdown characteristics of the pseudo-vertical GaN-on-Si p–n diode.

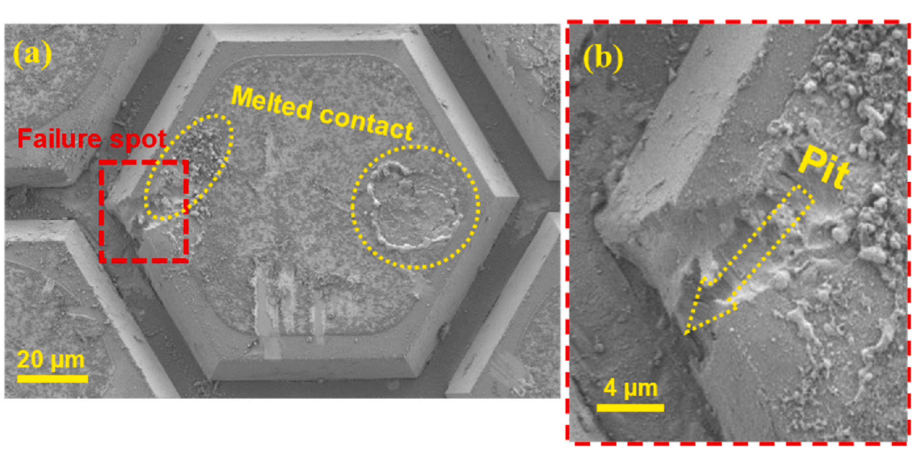

Fig. 12. Post-failure SEM analysis of the GaN diode: (a) top-view image showing surface damage, with (b) a magnified view near the anode edge (highlighted by a red rectangle). The yellow circle indicates the region of melted contact. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

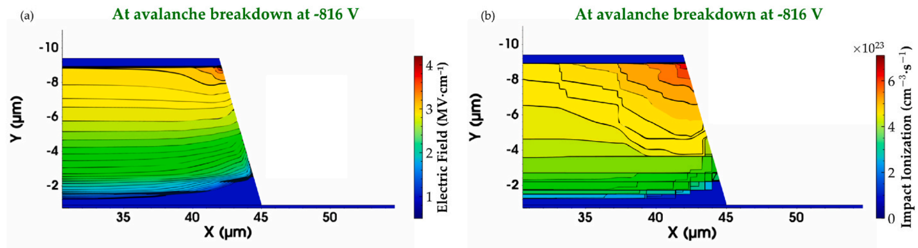

Fig. 14. (a) Simulated 2D electric field (EF) distribution at avalanche breakdown under a reverse bias of −816 V. (b) Simulated 2D impact ionization profile highlighting the onset of avalanche at the edge of the p–n junction.

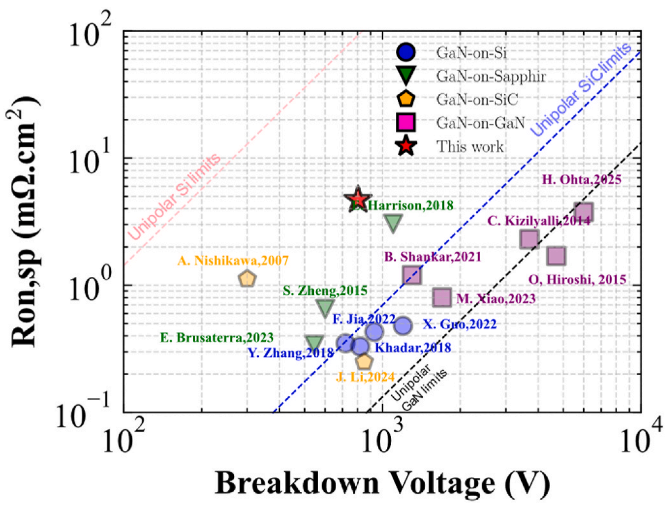

Fig. 15. Benchmark comparison of the p–n diode demonstrated in this work against state-of-the-art vertical GaN diodes on various substrates, including the unipolar limits of Si, SiC, and GaN.

ionization coefficients were used in the simulation. The simulated breakdown characteristics exhibit very close agreement with the measured J_R–V data, thereby confirming the validity and accuracy of the simulation model. The experimental results reach 98% of the TCAD-simulated breakdown voltage of 816 V, using the same bevel angle and doping conditions. It is important to note that leakage mechanisms, such as variable-range hopping (VRH), were explicitly incorporated into the simulation, as discussed previously in our prior work [26]. These extracted VRH parameters discussed above were subsequently implemented in the simulation framework to ensure accurate representation of the leakage behavior under the tested operating conditions.

The simulations further confirm that avalanche initiation occurs at the device edge—precisely the location identified in SEM observations—driven by strong electric field crowding. This is visualized in the 2D electric field distribution and impact ionization coefficient maps shown in Fig. 14(a) and (b), where peak field intensities coincide with the regions of observed physical damage.

Taken together, the experimental and simulation results demonstrate that insufficient edge termination leads to severe field non-uniformity, localized Joule heating, and eventual thermo-mechanical failure. Implementing robust termination structures is therefore essential to redistribute the electric field, reduce current crowding, mitigate localized thermal stress, and enhance device reliability under high-field and surge-current conditions.

Additionally, the p–n diode presented in this work is compared with state-of-the-art vertical GaN-on-GaN, GaN-on-Si, and SiC diodes in terms of specific on-resistance versus breakdown voltage, benchmarked against the unipolar limits of GaN, SiC, and Si, as summarized in Fig. 15. The device performance could be further improved by optimizing its geometry, reducing the specific on-resistance, and incorporating an effective edge-termination design.

4. Conclusion

In this work, we have demonstrated avalanche breakdown behavior at 805 V in vertical GaN p–n diodes grown by selective area growth (SAG) on Si, offering the advantages of lower cost and compatibility with standard silicon fabrication processes. Achieving high forward current and low specific on-resistance requires careful optimization of the device geometry and formation of high-quality ohmic contacts on the p-GaN layer.

The reverse-bias behavior was systematically examined using temperature-dependent leakage measurements and, for the first time in GaN devices, the DC stress method. This approach provides a highly controlled and reproducible way to probe avalanche behavior under steady or quasi-steady high-field operation. When biased slightly above the avalanche current, the diodes maintained stable avalanche operation for up to 7 min. Pre- and post-stress I–V characterization, together with SEM imaging, revealed that any degradation was localized near the periphery of the active area.

TCAD simulations, incorporating leakage mechanisms such as variable-range hopping (VRH), successfully reproduced the measured breakdown characteristics and confirmed that avalanche initiation occurs at the device edges.

Despite these advances, SAG is still under development, and larger mesas (>100 μm) with thicknesses exceeding 7 μm face challenges such as delamination and cracking. Therefore, further optimization of growth conditions and implementation of improved stress-management strategies are essential to maintain consistent material quality across different device geometries. Moreover, the development of advanced edge-termination schemes will be crucial for fully realizing the reliability potential of GaN power devices.

CRediT authorship contribution statement

Mohammed El Amrani: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **David Plaza Arguello:** Resources. **Zakariae M'Qaddem:** Resources, Formal analysis, Data curation. **Hala El Rammouz:** Resources. **Thomas Kaltsounis:** Resources. **Matthew Charles:** Writing – review & editing, Validation, Resources, Project administration, Investigation, Funding acquisition. **Daniel Alquier:** Writing – review & editing, Validation, Supervision, Investigation. **Julien Buckley:** Writing – review & editing, Visualization, Validation, Supervision, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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